# PARSA MADINEI

madinei@ucsb.edu | Website | LinkedIn | GitHub

5th-year PhD student working on visual perception in Vision & Image Understanding Lab at UC Santa Barbara. First-authored publications at CVPRF, IEEE SMC, and Journal of Vision. Experienced in building and shipping scalable ML systems using PyTorch and Huggingface.

## EDUCATION

**University of California, Santa Barbara** — Santa Barbara, CA

*Ph.D. Psychological & Brain Sciences (Visual Perception), GPA: 4.00* — *Expected 2027*

*M.Sc. Computer Science, GPA: 3.83* — *2023 - 2026*

**Coursework:** Adversarial Machine Learning, Advanced Computer Vision, Statistical Geometric ML, Scalable Internet Services, Human-AI Interaction, Computational Neuroscience, Runtime Systems

**University of Tehran** — Tehran, Iran

*B.Sc. Computer Engineering, GPA: 3.92 | B.Sc. Biology, GPA: 3.52* — *2016 – 2021*

## SELECTED PUBLICATIONS

- **Madinei, P.**, Solgi, R., Wen, Z., Skaza, J., Eckstein, M., Pedarsani, R. (2026). INTERLACE: Interleaved Layer Pruning and Efficient Adaptation in Large Vision-Language Models. CVPRF 2026.
- **Madinei, P.**, Karmakar, S., Hoffing, R., Gervits, F., Eckstein, M. (2026). IRIS: Intent Resolution via Inference-time Saccades for Open-Ended VQA in Large Vision-Language Models. IEEE SMC 2026 Submission.
- Wen, Z., **Madinei, P.**, Eckstein, M. (2026). Revealing the Gap in Human and VLM Scene Perception through Counterfactual Semantic Saliency. NeurIPS 2026 Submission.
- Solgi, R., **Madinei, P.**, Tian, J., Swaminathan, R., et al. (2026). Activation-Informed Pareto-Guided Low-Rank Compression for Efficient LLM/VLM. EMNLP 2027 Submission.
- **Madinei, P.**, Shehabi, S., Eckstein, M. (2026). Feature-Based Language Guidance Facilitates Visual Search in Humans and Foveated Vision-Language Transformer Models. Journal of Vision, 2026.
- Skaza, J., **Madinei, P.**, Wen, Z., Eckstein, M. (2025). DReX: Pure Vision Fusion of Self-Supervised and Convolutional Representations for Image Complexity Prediction. arXiv preprint.
- **Madinei, P.**, Eckstein, M. (2025). Language Guided Search with Multi-Modal Foveated Deep Neural Network Finds Objects at Unexpected Locations, Journal of Vision, 2025

## RESEARCH EXPERIENCE

**Graduate Research Assistant — Vision & Image Understanding Lab, UCSB** — Mar 2023 – Present

- Developed INTERLACE, a model compression framework combining interleaved layer pruning with parameter-efficient fine-tuning for large vision-language models, reducing model size while preserving accuracy across diverse benchmarks.
- Designed novel vision-language systems (VLMs/LLMs) integrating real-time eye-tracking data to resolve ambiguity in visual question answering (VQA), achieving state-of-the-art results on multiple benchmarks.
- Built end-to-end deep learning pipelines in PyTorch for large-scale multimodal experiments; managed data preprocessing, distributed training (DeepSpeed), and evaluation across compute clusters.
- Collaborated cross-functionally with Cognitive Science, engineering, and HCI teams to design human studies (IRB-approved) and deploy ML systems for real-time human-AI interaction research.

**Google CSRMP Researcher** — Feb 2023 – May 2023

- Conducted computer vision research under mentorship of Google Research scientist Thomas Mensink; gained hands-on experience with state-of-the-art pattern recognition and efficient learning techniques.

## TECHNICAL SKILLS

**Languages:** Python, C/C++, R **Frameworks:** PyTorch, HuggingFace Transformers, DeepSpeed, Torchvision
**Domains:** Vision-Language Models (VLMs), Computer Vision, LLMs, NLP, Machine Learning, Data Mining
**Tools & Libraries:** Git, Linux, AWS, CUDA, OpenCV, Scikit-learn, NumPy, SciPy, Pandas, Plotly, Matplotlib, Weights & Biases

## PRESENTATIONS & SERVICE

- Awarded Talk, Vision Sciences Society (VSS) Conference (2025): Presented multimodal foveated transformer model for language-guided visual search.
- UCSB's "Beyond Academia" Organizing Team: Developed teamwork and leadership skills through collaboration with the organizing team to organize career panels. Utilized coaching and communication skills to help attendees create personalized action plans for their careers
- Graduate Teaching Assistant: AI & Visual Cognition, Perception, Lab in Visual Perception, Complex Systems Analysis with Mathematica, Visual Neuroscience, Statistical Research Methods (UCSB, 2022–Present).